

lec 8

0, -7

Lec_8

lec 9

20-30

Practice Questions on time
Complexity

1-19

20.

compare

Main()

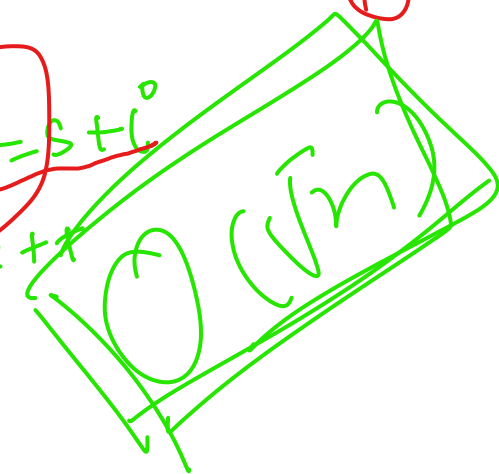
{ s=0, i=0;

While(s<=n)

{ s=s+i

i=i+1

s = s + i



$$s=0, i=0, s<=n, n=10$$

$$0+0$$

$$1$$

$$s = s + i$$

$$10$$

$$i = i + 1$$

$$0 < 10,$$

$$s = 0 + 0,$$

$$i = 1$$

$$0 < 10,$$

$$s = 0 + 1$$

$$i = 2$$

$$1 < 10,$$

$$s = 0 + 1 + 2$$

$$i = 3$$

$$3 < 10,$$

$$s = 0 + 1 + 2 + 3$$

$$i = 4$$

$$6 < 10,$$

$$s = 0 + 1 + 2 + 3 + 4, i = 5$$

$$s = 1 + 2 + 3 + 4 \dots$$

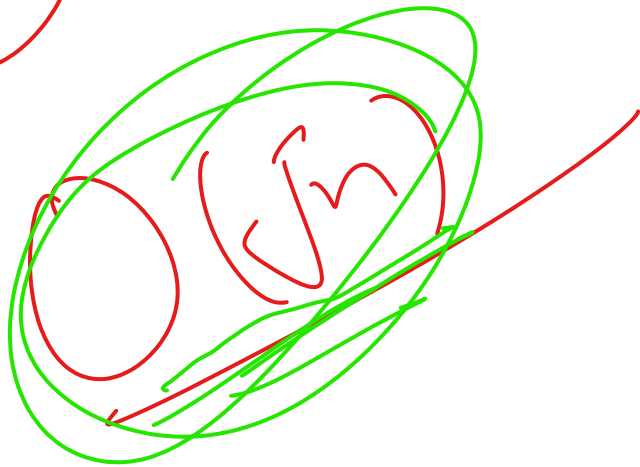
$$s = 1 + 2 + 3 + 4 + 5 \dots k-1 +$$

$$s <= n$$

$$\text{sum of first } n \text{ numbers} = \frac{n(n+1)}{2}$$

$$k = n$$

$$\Rightarrow s = \frac{k(k+1)}{2}$$



$$s \geq n$$

$$\Rightarrow \frac{k(k+1)}{2} = n$$

$$\Rightarrow \frac{k^2 + k}{2} = n$$

$$\Rightarrow k^2 = n$$
$$k = \sqrt{n}$$

$$O(k^2 + k)$$

$$= O(k^2)$$

$$O(n^2 + n + 1)$$
$$= O(n^2)$$

21.)

Main()

```
{  
  for(i=1; i<=n; i++)  
  {  
    for(j=1; j<=n; j++)  
    {  
      for(k=1; k<=n; k++)  
      {  
        x=y+z  
      }  
    }  
  }  
}
```

Loop 1

Loop 2

Loop 3

inside 1

inside 2

i=1, i=2, i=3, i=4, ..., i=n.

$T_c = n \times \text{inside 1}$

$T_c = n \times n \times \text{inside 2}$

inside = 3

J=1, J=2, J=3, ..., J=n

$T_c = n \times n \times n$
 $= n^3$

$O(n^3)$

~~$O(n)$~~
 ~~$O(n)$~~
 ~~$O(n)$~~

~~for $i=1; i \leq n; i++$~~
 ~~$n = n + 2$~~

for $j=1; j \leq n; j++$

$n = n + 2$

for $k=1; k \leq n; k++$
 $n = n + 2$

n

n

n

22.)

Main()

```
{  
  for(i=11; i<=n^5; i++)  
  {  
    for(j=199; j<=n^78; j++)  
    {  
      for(k=21; k<=n^11; k++)  
      {  
        x=y+z;  
      }  
    }  
  }  
}
```

11 to n^5

$(n^5 - 11)$ times

1 loop $\Rightarrow$ n times

2 loop $\Rightarrow$ n-1 times

3 loop $\Rightarrow$ n-2 times

4 loop $\Rightarrow$ n-11 times

$O(n^9)$

$(n^{78} - 199)$ times

$T(C) = (n^5 - 11) \times \text{inside 1}$

$T(C) = (n^5 - 11) \times (n^{78} - 199) \times \text{inside 2}$

$T(C) = \frac{(n^5 - 11) (n^{78} - 199) (n^{11} - 21)}{n^5 \times n^{78} \times n^{11}}$

$T(C) = \frac{n^{93} \times n^{11}}{n^{94}}$

inside 2 inside 4

23.)

Main()

{

for(i=11;i<=n^5/23;i++)

{

for(j=199;j<=n^78/56;j++)

{

for(k=21;k<=n^11/3;k++)

{ x=y+z

}

}

}

}

23

homework

$$i <= n^5/23$$

$$j <= n^{78}/56$$

$$k <= n^{11}/3$$

Comment 2 lines

24.)

Main()

{

for(i=1; i⁴ ≤ n²⁵⁶; i++)

{

for(j=1; j³ ≤ n²⁷; j++)

{ x=y+z;

}

}

}

n⁴ × inside
n⁴ × n³
-- O(n⁷)

34

i⁴ ≤ n²⁵⁶

35

$$i=1,$$

$$i=2,$$

$$i=3,$$

$$i=4,$$

$$\begin{cases} i^4 \leq n^{256}, & i=2 \\ i^4 \leq n^{256}, & i=3 \\ i^4 \leq n^{256}, & i=4 \\ i^4 \leq n^{256}, & i=5 \end{cases}$$

$$n = n^4$$

$$n^4 \geq n^{256}$$

$$n^4 = n^{256}$$

$$n^2 = n^6$$

$$n = n^4$$

$$n^3 = n^{27}$$

$$(n^3)^{1/3} = (n^{27})^{1/3}$$

$$n = n^3$$

